
Education

University of Utah*B.S. Computer Science***Salt Lake City, UT***August 2020 – May 2025*

Relevant Coursework: Computer Systems, Machine Learning, Computer Graphics, Algorithms, Software Practice I & II, Database Systems, Computer Networks, Foundations of Data Analysis, Models of Computation, and Linear Algebra.

Skills

Languages: Python, TypeScript, C++, SQL, Java

Software: Pandas, NumPy, React, Node.js, AWS, Azure, Docker, Next.js, .NET, REST, Django, MongoDB, Linux

Experience

L3Harris Technologies*Associate Software Engineer***Dallas, TX***June 2025 – Present*

- Improved sensor data throughput and latency on defense embedded platforms by developing real-time signal processing algorithms in modern C++ on embedded Linux, applying system-level profiling and deterministic scheduling to meet strict hardware-in-the-loop constraints.
- Reduced deployment risk across multiple production targets by building CI/CD pipelines with static analysis, integration testing, and automated validation, enabling reliable releases to embedded compute platforms in collaboration with hardware and firmware teams.
- Increased data pipeline reliability for high-frequency sensor ingestion by engineering containerized microservices with fault isolation and recovery patterns, sustaining throughput under resource-constrained production conditions.

Doxy.me*Software Engineer***Charleston, SC***August 2024 – May 2025*

- Improved clinical diagnostic accuracy by building and deploying end-to-end ML inference pipelines in Python and TypeScript on AWS SageMaker and Fargate, owning observability, latency optimization, and reinforcement learning feedback loops across production services.
- Reduced integration latency for real-time video workflows by integrating WebRTC streaming directly into the inference loop, enabling low-latency data flow from live clinical sessions to model consumers.
- Enabled zero-downtime releases across multiple production environments by owning YAML-based CI/CD configuration and containerized infrastructure, cutting environment drift and release cycle time.

University of Utah*Undergraduate Research Assistant – FuTURES Lab***Salt Lake City, UT***May 2024 – Jun 2025*

- Analyzed and optimized large-scale datasets, enhancing software configuration testing with gcov and CMake, increasing code coverage by 30% on real-world APIs (Libpng, PyTorch).
 - Expanded OSS-Fuzz testing coverage using compile-time options to improve the robustness of full-stack libraries.
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Projects

Configuration Fuzzer: Developed a configuration fuzzing tool for OSS-Fuzz, automating build generation to identify critical compile-time configurations and improve code coverage, uncovering previously untested code paths and increasing bug detection effectiveness.

Telehealth Mental Health Diagnosis Tool (Doxy.me): Built an AI system that analyzes visual, speech, and tonal cues from telehealth sessions using PyTorch and Llama, delivering structured mental health insights to treating physicians via SageMaker and Fargate inference pipelines. Improved diagnostic accuracy using reinforcement learning and WebRTC video integration.

University Course Planning/Review Platform (Class Best): Built a multi-service data platform enabling student schedule matching, course overlap detection, and behavioral insights. Developed ETL pipelines in Python to process thousands of course records, user preferences, and time-series activity logs. Built a responsive frontend in React + Tailwind with interactive dashboards, heatmaps, and calendar visualizations.